Linc(d) Notebook System Spec (Prototype v0)

1. Overview
The Linc(d) Notebook is a unified cognitive workspace where multimodal inputs (text, images, audio, data, tools) are ingested, structured, and compiled into multiple output forms such as documents, slideshows, videos, dashboards, and conversational agents.

It functions as a format-agnostic cognitive compiler:
Input → Context Graph → Representation Layer → Output Artifact(s)

---

2. Core Input Types

2.1 Text Inputs
- Notes
- Conversations
- Prompts
- Research dumps
- Structured markdown

2.2 Media Inputs
- Images (photos, diagrams, UI mockups)
- Audio (voice notes, interviews)
- Video clips
- Screenshots

2.3 Data Inputs
- CSV / tables
- JSON logs
- Sensor streams
- App usage data

2.4 Tool Inputs
- API calls
- MCP tools
- Web queries
- Local system commands

---

3. Internal Representation

3.1 Unified Context Graph
All inputs become nodes:

- Entity Nodes (people, places, concepts)
- Event Nodes (things that happened)
- Artifact Nodes (files, images, outputs)
- Intent Nodes (goals, tasks)

Edges define:
- causality
- similarity
- sequence
- relevance

---

4. Output Compilation Layer

The notebook can compile the same source context into multiple formats:

4.1 Text Report
- structured markdown or PDF
- white paper style output
- technical documentation

4.2 Slide Deck
- auto-generated presentation frames
- each node becomes a slide
- emphasis on hierarchy + visuals

4.3 Video Script / Generation Feed
- scene-based breakdown
- narration text
- optional image prompts per scene

4.4 Podcast Mode
- conversational script between agents
- structured dialogue from context graph
- tone-controlled narration

4.5 Dashboard Mode

- real-time widgets
- memory graph visualization
- task + context overlays

---

5. LLM Orchestration Layer

The system uses multiple LLM roles:

5.1 Parser Model
- converts raw input → structured nodes

5.2 Reasoning Model
- interprets relationships
- maintains coherence across time

5.3 Compiler Model
- converts graph → output formats

5.4 Critic Model
- validates consistency
- removes hallucinated links
- enforces structure integrity

---

6. Example Notebook Session

User prompt:
"Plan a weekend with dog activity, art gallery, and rest time."

Process:
1. Extract intents:
   - recreation
   - pet activity
   - cultural exposure
   - recovery

2. Build graph:
   - Oliver (dog)
   - art gallery visit

   - rest block
   - energy sequencing

3. Compile outputs:

Text:
- structured itinerary

Slide deck:
- overview
- morning activity
- gallery visit
- rest period

Podcast:
- dialogue planning session

Dashboard:
- timeline + energy curve

---

7. Key Principle

The notebook is not a document.

It is a live compiler of cognition into media formats.

---

8. Extensions

- AR contextual overlays
- continuous memory sync
- multi-user shared graphs
- AI-to-AI publishing layer
- life-stream ingestion

---

9. Simple Minds Studios Layer

A creative intelligence studio layer that:
- transforms thought into structured media
- bridges intelligence and design
- acts as publishing layer for Linc(d)